



โจทย์ DragonAttack

In a mysterious valley, an ancient dragon is protecting a valuable treasure. The dragon has N body parts arranged in a line. Each body part has its own amount of HP (health points).

A warrior has a weapon with an attack power K . Each strike can attack up to M consecutive body parts. For every strike:

1. The warrior chooses a body part.
2. The weapon attacks that body part and up to the next $M - 1$ consecutive body parts.
3. Every body part hit by the strike loses exactly K HP.
4. If a body part's HP becomes less than 0, its HP is changed to 0.
5. A body part with 0 HP is considered destroyed. Further attacks have no effect on it.

The warrior may choose a different starting position for every strike.

Your task is to determine the minimum number of strikes needed to destroy all of the dragon's body parts.

Additionally, after every strike, output the current HP of all body parts.

Input

The first line contains three integers:

N, K, M - where:

- N = number of body parts
- K = attack power of the weapon
- M = maximum number of consecutive body parts that can be hit by one strike

The second line contains N integers: $H_1 H_2 H_3 \dots H_N$ - where H_i represents the initial HP of body part i .

Constraints

$$1 \leq M \leq N \leq 100,000$$

$$1 \leq K \leq 1,000$$

$$1 \leq H_i \leq 1,000,000$$

Output

After every strike, output:

Strike X : $H_1 H_2 H_3 \dots H_N$

where X is the strike number and H_i is the current HP of each body part after that strike.

After all body parts have been destroyed, output:

Total strikes: X - where X is the minimum number of strikes used.

TESTCASE

input	output
5 3 2 10 5 7 2 8	Strike 1: 7 2 7 2 8 Strike 2: 4 0 7 2 8 Strike 3: 1 0 7 2 8 Strike 4: 0 0 7 2 8 Strike 5: 0 0 4 0 8 Strike 6: 0 0 1 0 8 Strike 7: 0 0 0 0 8 Strike 8: 0 0 0 0 5 Strike 9: 0 0 0 0 2 Strike 10: 0 0 0 0 0 Total strikes: 10
4 3 2 5 2 4 1	Strike 1: 2 0 4 1 Strike 2: 0 0 1 0 Strike 3: 0 0 0 0 Total strikes: 3
3 2 2 4 3 2	Strike 1: 2 1 2 Strike 2: 0 0 0 Total strikes: 2